

Ring Galaxy Detection in Euclid Images Using Convolutional Neural Network and ResNet50

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PHYS 310 Final Report

1. Introduction

Ring galaxies are rare galaxies that have a bright center, like many other galaxies, but they have a bright ring of stars circling them far from the center [6]. They likely form when one galaxy collides directly with another, sending a wave outward that creates a ring of new stars. Studying them helps us understand how galaxies interact and change over time. In the past, galaxy shapes were classified manually by people looking at images. The Galaxy Zoo project [1] used many volunteers to label galaxies. These labels are values between 0 and 1, derived from how many people voted yes or no to the image. The Euclid space telescope [2] is a recent mission that will image billions of galaxies. Their images are high quality, but the image dataset is huge to check manually. In this project, a convolutional neural network (CNN) is trained on Galaxy Zoo data to predict ring galaxy probability. The model is then applied to Euclid images, where it predicts high-probability ring galaxies.

2. Methods

2.1 Dataset

Two datasets are used:

- `gz_rings` (Walmsley et al., [3]): A labeled dataset of about 91,000 galaxy images from Galaxy Zoo, split into a training set (73,356 images) and a test set (18,340 images). Each image is labeled with a ring probability, determined by the number of votes of the volunteers. The label distribution is heavily skewed toward 0, as ring galaxies are rare.

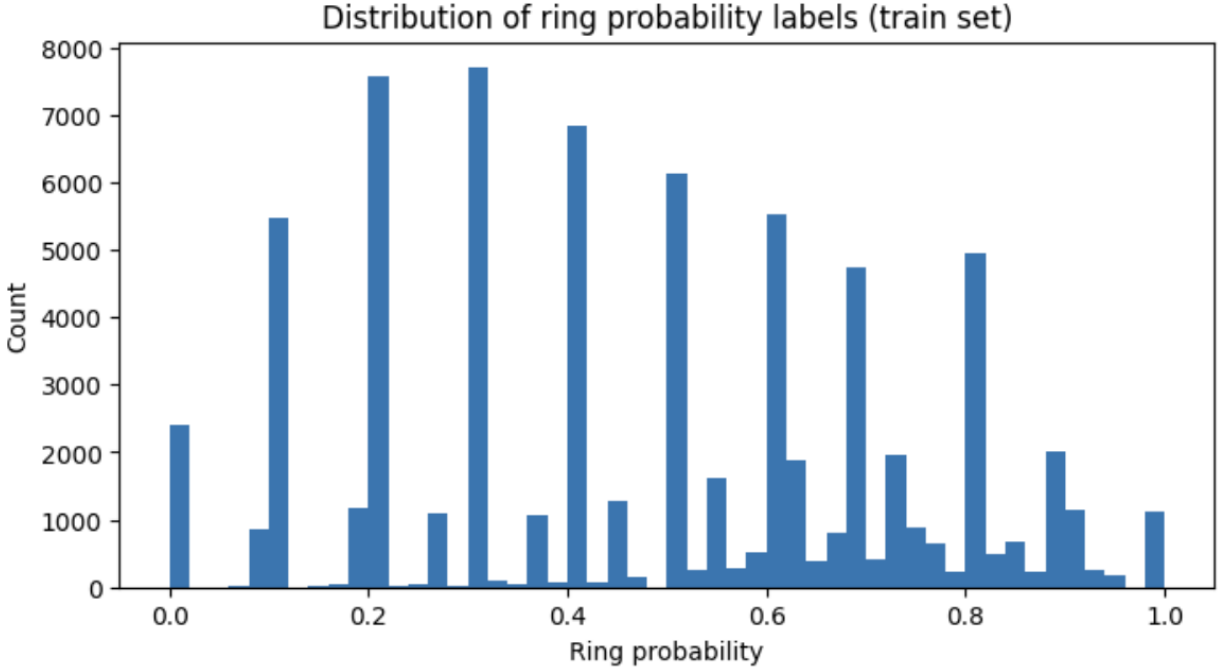


Figure 1: The distribution of the probability of ring galaxies. Smaller probabilities are shown to have higher counts.

- `gz_euclid` (Walmsley et al., [4]): A dataset of galaxy images from the Euclid telescope, containing 2,378 images in its test split, which are used for the model's prediction. These images have no ring probability labels, so the trained model is used to infer their predicted probability and rank them.

2.2 Model Architecture

The model uses ResNet50 [5], a deep CNN pretrained on ImageNet. The pretrained backbone provides general visual features (edges, curves, textures) that are also useful for galaxy images, without requiring training from scratch. The classification layer of ResNet50 is removed and replaced by a new layer, consisting of:

- a global average pooling layer (compresses the backbone output into a flat vector of 2048 numbers)
- two dropout layers (randomly disables 30% of neurons during training to reduce overfitting)
- a dense layer with 128 units and ReLU activation
- a single output neuron with sigmoid activation (the sigmoid activation limits the output to the range 0 to 1 only, which is what we want for the probability labels).

ResNet50 has 175 layers. The first 145 layers are frozen (their pretrained weights stay fixed during the training). Only the last 30 layers are allowed to update during training because those last layers of ResNet50 are highly responsible for learning pattern like ring-shape, allowing the model to learn from the new galaxy images.

2.3 Training

The model is trained using: Adam optimizer with learning rate $1e-4$, Mean Squared Error (MSE) loss, and Mean Absolute Error (MAE) as a secondary metric. To manage 8Gb memory constraints, instead of loading all 72,000 training images into RAM simultaneously, a lazy tf.data pipeline is used, where images are decoded and preprocessed one at a time via a Python generator. Only a part of the Galaxy Zoo dataset is used due to running time and computing constraint in my M1 chip MacBook (notes that the dataset was shuffled before being selected to reduce bias):

- 20,000 samples for training
- 4,000 for validation
- 4,000 for final testing

The model updates the weights iteratively with a batch of 32 images until it reaches 20,000 images per epoch and finishes after 13 epochs.

2.4 Preprocessing

All images are resized to 224×224 pixels, matching the input size of ResNet50. Images are preprocessed using ResNet50's standard preprocessing function, which subtracts ImageNet channel means and scales pixel values appropriately. This step is critical. Using simple normalization to $[0, 1]$ leads to a mismatch with pretrained weights and prevents effective learning.

3. Results

3.1 Training Curves

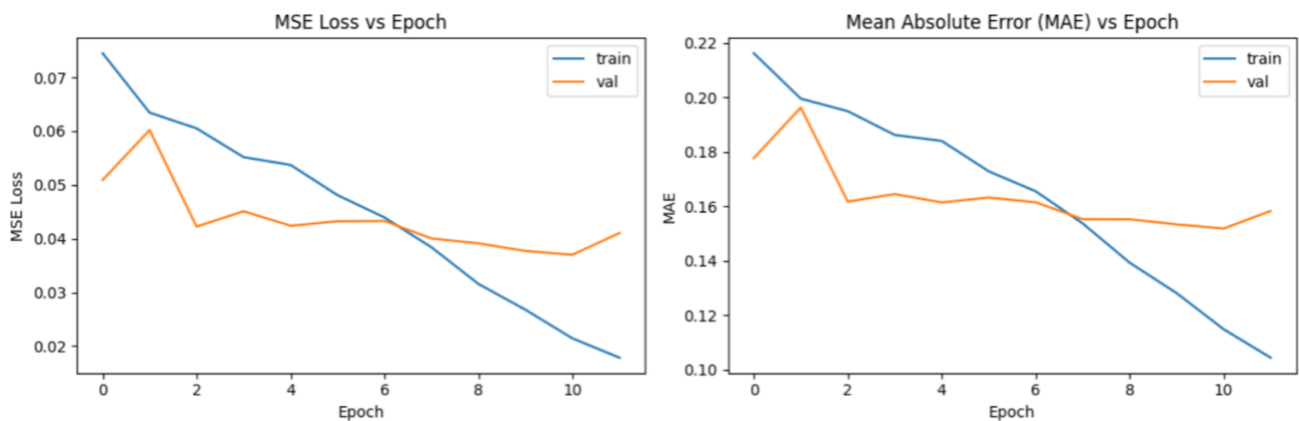


Figure 2 and 3: MSE Loss and MAE loss across epochs of train and validation sets.

During the initial epochs, both train loss and validation loss decreased (with training loss decreased slightly more). However, after the sixth epoch, the train loss kept decreasing with

similar rate, but the validation loss plateaued around 0.040. This suggests a sign of mild to moderate overfitting, and I should have stopped at around epoch 6 to 7.

3.2 Test Set Evaluation

The model was evaluated on 4,000 test images from `gz_rings`, yielding a test MSE of 0.042 and test MAE of 0.1563. These scores fall between the MSE and MAE scores of the train and validation set, indicating that the models generalized quite well and the overfitting issue was not too severe.

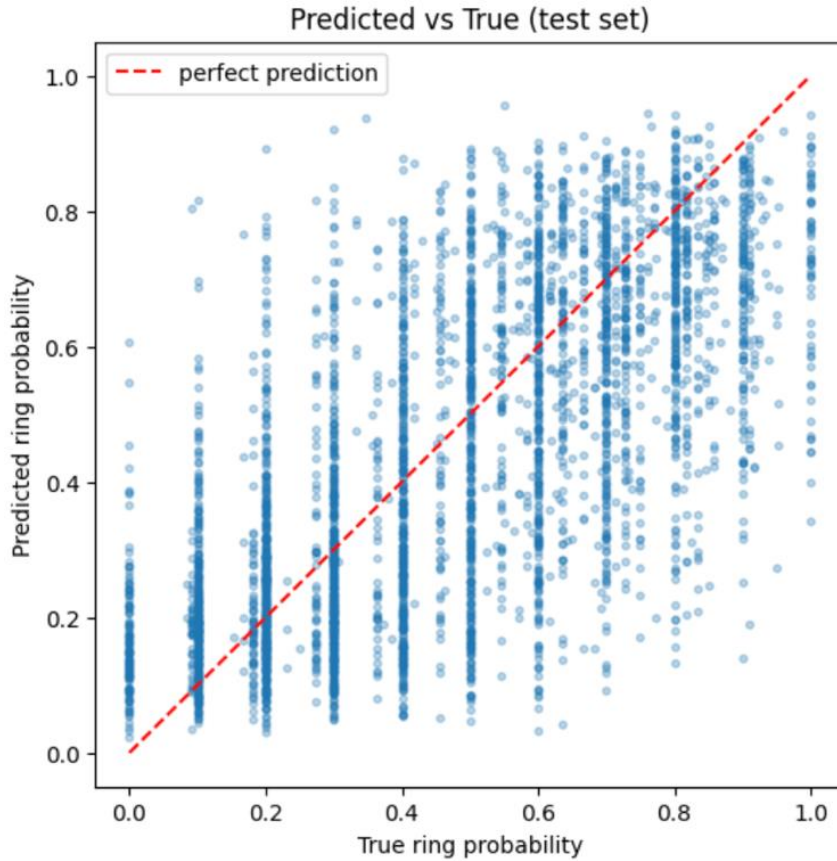


Figure 4: The scatter plot of true probability labels vs predicted probability labels showing a noticeable linear relationship.

We can see a linear relationship from the above scatter plot; this suggests that the model can generalize and predict on the new test set quite well. However, there is a large vertical spread at every true probability value, meaning prediction precision is low. The model performs more reliably at high and low ring probabilities (below 0.3 and above 0.6), where points cluster closer to the red line. Overall, the model captures the general trend but has limited precision, especially for the galaxies with a probability label between 0.3 and 0.6.

3.3 Inference on Euclid Images

The trained model was applied to 2,378 Euclid images. Each image was assigned a predicted ring probability and ranked accordingly.



Figure 5: Top 25 highest probability images predicted by the model from the Euclid dataset.

From the predicted images above, we can see that most candidates display a visible ring-like structure. However, some of them might be false positives. For example, there is not enough evidence to tell whether galaxy rank 6 is a ring galaxy or not, as the angle of capturing is mostly flat. In addition, galaxy 17 can be a spiral galaxy since I can see its arms. These two likely false

positives are expected given the domain shift between training (Galaxy Zoo/SDSS) and Euclid images.

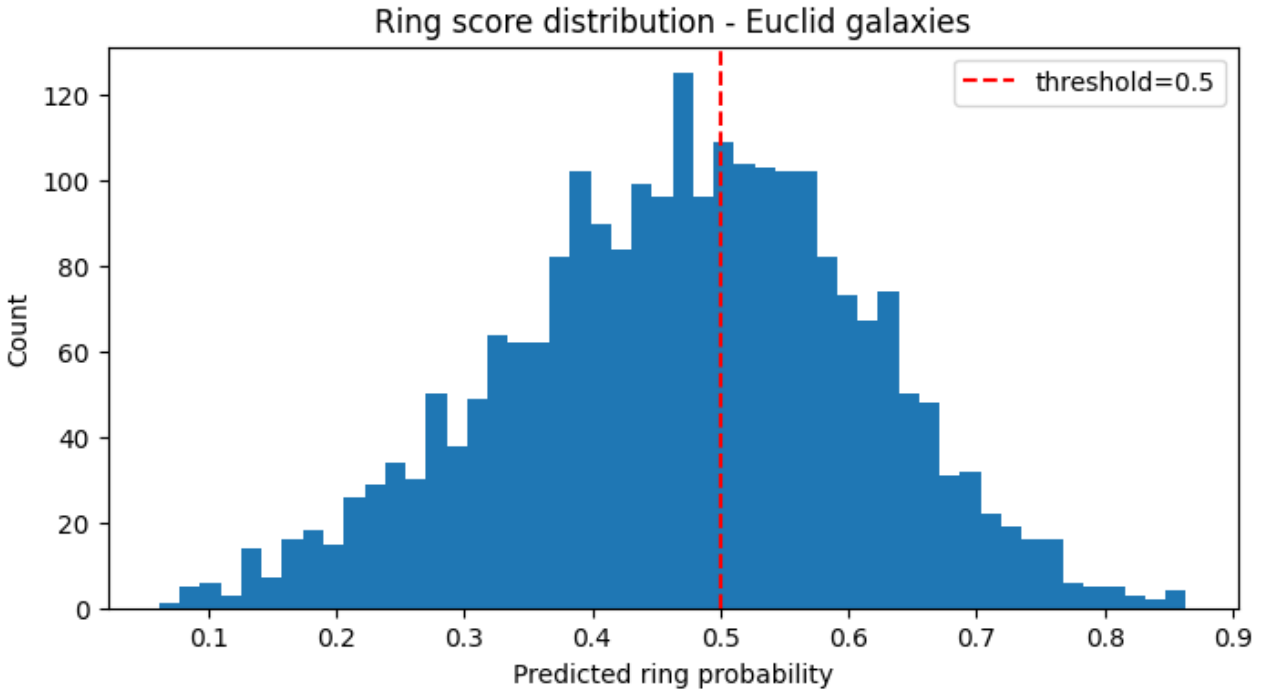


Figure 6: Histogram describing the model's predicted probability distribution

From the histogram, it shows that the distribution across all 2,378 Euclid images is approximately bell-shaped, centered near 0.5, with scores ranging from ~ 0.07 to ~ 0.88 . Also, there are 1044 galaxies scored above the 0.5 threshold. The wide distribution suggests the model is differentiating between galaxies rather than just predicting a constant value. Though the center near 0.5 suggests that the model was uncertain of many predictions - doesn't know if an image is a ring galaxy or not. This is expected too, given the fact that the model was trained on Galaxy Zoo images from the SDSS telescope (older telescope, image quality, color, and resolution), and applied to predict Euclid images (different telescope, different camera, sharper resolution, and different brightness/contrast).

4. Discussion

The model successfully learned to predict ring galaxy probability from Galaxy Zoo labels. On the test set, it achieved an MAE of 0.156, and the scatter plot shows a clear positive trend between predicted and true probabilities. Applied to 2,378 Euclid images, the model identified 1,044 candidates scoring above 0.5, with most of the top-ranked images visually showing ring-like structures. In addition, several challenges and limitations were encountered during the project, some of which need future work:

- Memory and computation limitations: Loading all training images into RAM at once crashed the kernel. A lazy data pipeline was used to load images one at a time. The same

issue occurred with Euclid data, limiting inference to 2,378 images instead of the full 169,899. Additionally, the M1 MacBook's limited GPU made each training run on the current dataset take around 90 minutes, making it impractical to use the full dataset or tune hyperparameters easily. With more RAM and a stronger GPU, the full Galaxy Zoo and Euclid datasets could be used, likely improving both model performance and prediction coverage.

- **Model not learning:** Two separate issues caused the model to predict a constant value. First, incorrect image preprocessing (normalizing to $[0,1]$ instead of using ResNet50's preprocessing function) caused a mismatch with pretrained weights. Second, calling the backbone with `training=False` prevented batch normalization layers from adapting to galaxy images. Both were identified and fixed.
- **Overfitting:** Validation loss plateaued around epoch 6 while training loss continued dropping. Early stopping should be applied in future runs.
- **Domain shift:** The model was trained on SDSS telescope images but applied to Euclid images, which have different resolution and contrast. This likely contributed to false positives in the top candidates, and the score distribution centered near 0.5. For future work, training the model on a small number of labeled Euclid images might improve the model's prediction on those images.
- **Labels of the train dataset:** Galaxy Zoo image labels are voted by volunteers, which might introduce much noise. This can be improved if the labels can only be voted on by professionals and/or undergraduate students in the field.

6. Conclusion

This project demonstrated that a pretrained ResNet50 can be fine-tuned to predict ring galaxy probability from crowd sourced Galaxy Zoo labels and applied to new telescope data. The model achieved a test MAE of 0.156 and generalized well to unseen data. Applied to 2,378 Euclid images, it identified 1,044 candidates scoring above 0.5, with most top-ranked images visually showing ring-like structures. The results show that transfer learning is an effective approach for galaxy morphology tasks, even with limited compute and memory. However, domain shift between training and Euclid images is a big obstacle to obtain higher prediction confidence. With more memory, the full 169,899 Euclid images could be processed. With more compute, hyperparameter tuning and early stopping could reduce overfitting. Training on a small set of labeled Euclid images could further improve results.

Code submitted as supplementary Jupyter notebook:
`ring_galaxy_project.ipynb`

7. References

- [1] Galaxy Zoo Project, Zooniverse. Available:
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https://www.esa.int/Science_Exploration/Space_Science/Euclid
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https://huggingface.co/datasets/mwalmsley/gz_rings
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- [5] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” 2016. Available: <https://arxiv.org/abs/1512.03385>
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